Worksheet LaTeX Guide

Worksheets for Student Services are created in LaTeX to allow for display of mathematical expressions and proper formatting. There are multiple ways to edit and compile .tex files, but the easiest way that requires the least set-up is through Overleaf. If you'd prefer to download a LaTeX distribution instead, refer to the readme file in the worksheets repository.

Grainger provides an Overleaf Premium subscription free of charge to all engineering students. More information about this can be found at this link. The important thing to note is that you must sign in through your University of Illinois account by clicking the "Log in with SSO" button.

Once you're logged in, you're ready to start working on worksheets. It is recommended to create a new project for each class or each worksheet. This will keep your files separated and easier to find when you go to edit them.

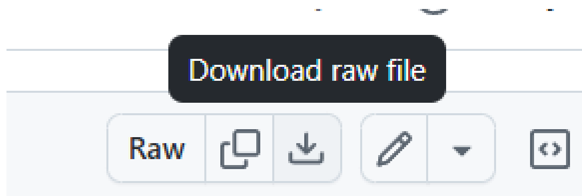
You are able to share each project to allow for collaboration, similar to Google Docs. This is extremely helpful, as worksheets can be a lot of work. Splitting the work between multiple partners is a good way to finish worksheets faster. Therefore, it is **highly** recommended to collaborate with at least one other person. Plus, you can check each other's work for correctness as you go. Initiates, this also means you can both claim service points.

Editing Existing Worksheets

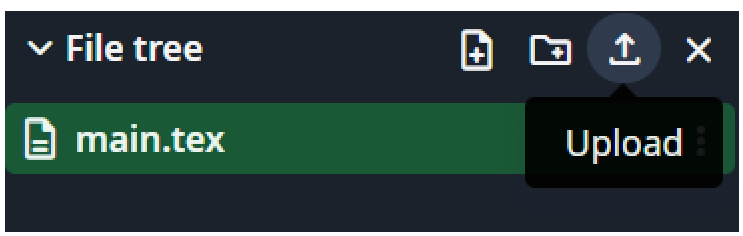
Each worksheet on GitHub will have four files:

1. worksheet.tex
2. worksheet.pdf
3. solutions.tex
4. solutions.pdf

The files you will need to edit are the worksheet.tex and solutions.tex files. You can download them by navigating to the file and clicking the download button in the top right corner.

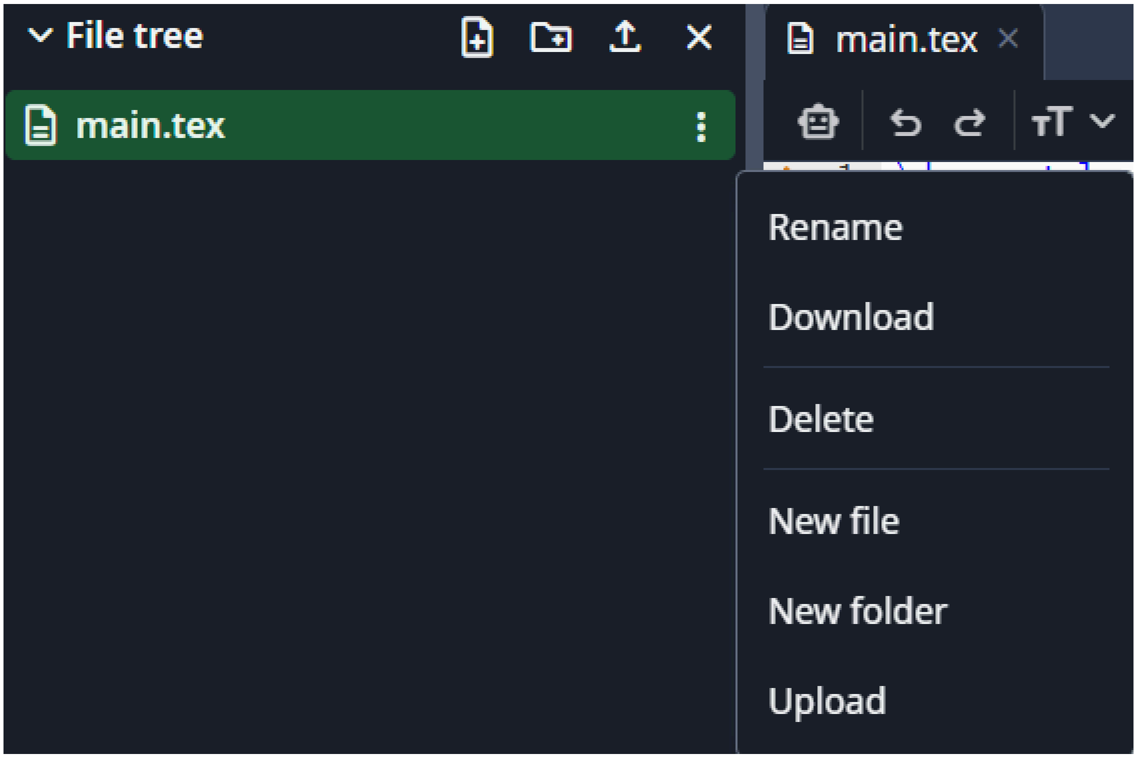


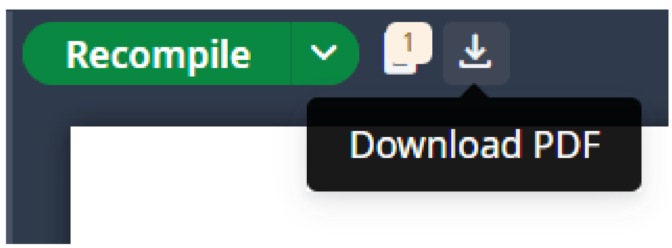
Once the files are downloaded, head over to Overleaf. Click the upload button in the top left corner and select both files. The main.tex file is automatically created when you create a project, you can safely delete this as it won't be needed.



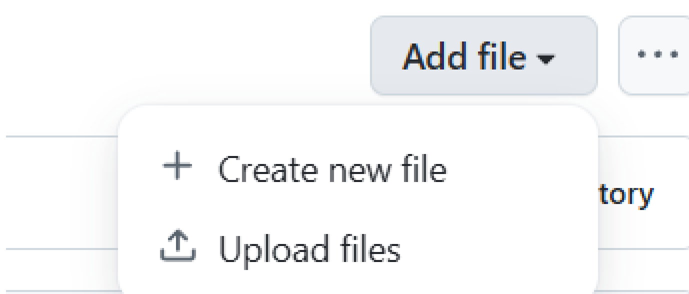
Once your files are uploaded, you can simply edit them in the left-hand window. Make sure to click the "Recompile" button on the right-hand window while editing. This will compile your .tex file and create a .pdf which you can view on the right. I recommend checking this often to confirm your worksheet is formatted correctly.

When you're completely done with the worksheet and solutions, download the .tex and .pdf files for both. The .tex files can be downloaded by clicking the three dots next to each file and selecting "Download". The .pdf files can be downloaded by clicking the download button next to "Recompile". Make sure you recompile between downloading the worksheet and solutions, Overleaf does not automatically do this when you switch .tex files.

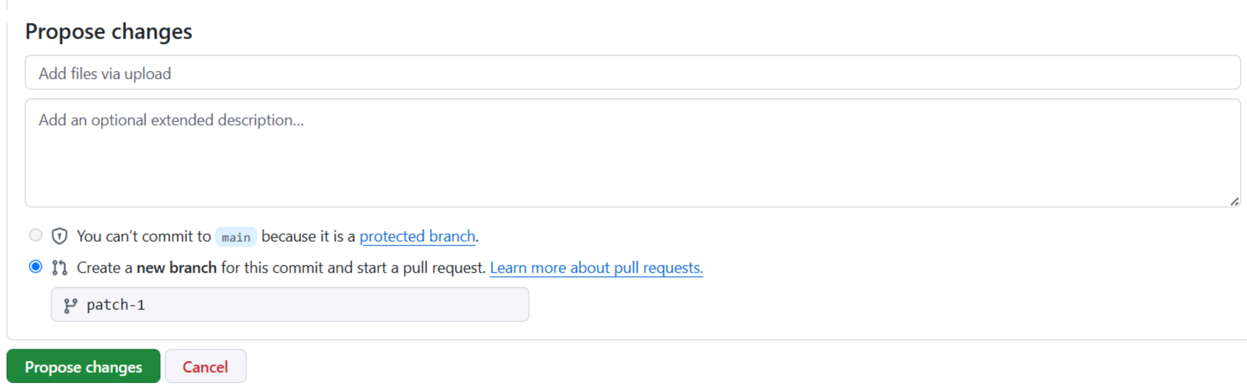




The .tex files should be named correctly already, but make sure you rename the .pdf files to “worksheet.pdf” and “solutions.pdf” respectively. Make sure not to mix up the file names, or it will be confusing for everyone. To upload your updated files to GitHub, go to the correct worksheet folder and select “Add file” and upload your files. Since your files should be named the same as the existing files, they will be replaced. Make sure this happens or there could be outdated worksheets or solutions.



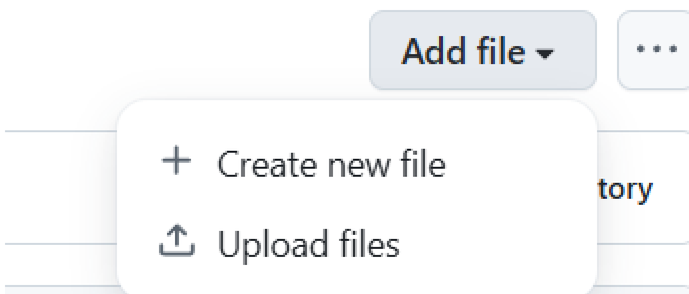
Once your files are uploaded, add a proper title and description. Create a new branch and start a pull request by clicking “Propose changes”. A Student Services Chair will review your changes and, if everything looks good, will merge your branch into main. Congratulations, you’ve helped Student Services improve the resources available to the ECE community at Illinois!



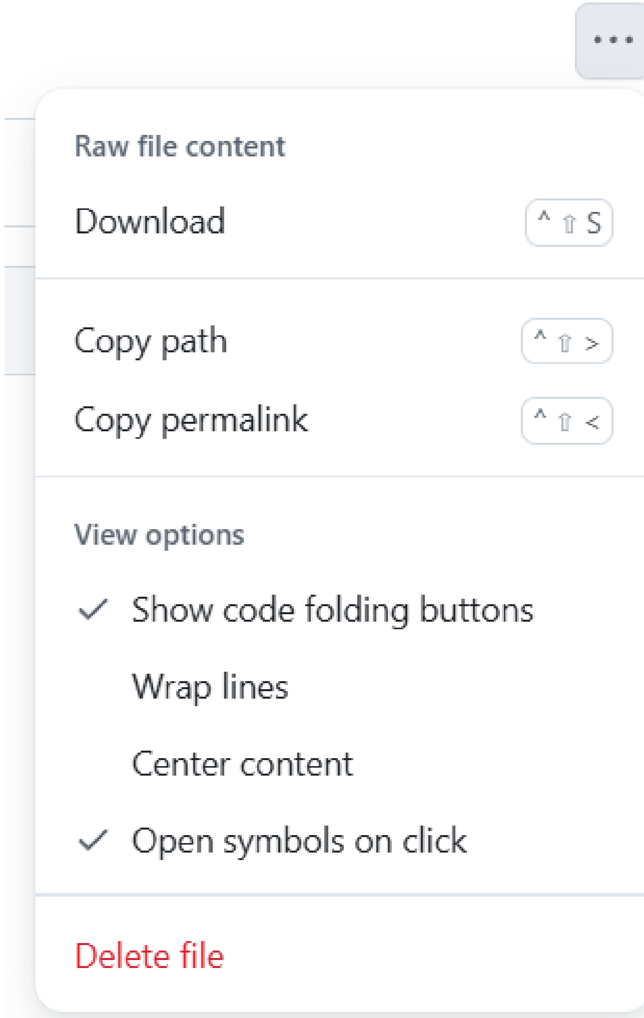
Creating New Worksheet

The process of creating a new worksheet is very similar to editing an existing worksheet. The only difference is that you may want to keep and rename the default main.tex file to “worksheet.tex” in order to save some time formatting your title page. Refer to the section above for how to compile your worksheet and solutions into .pdf files. Make sure you keep the file naming conventions the same as previously stated to avoid any conflicts when uploading to GitHub.

If there isn’t already a folder for the midterm that you created the worksheet for, you must create one. However, Git doesn’t allow you to create empty folders. So, first click “Add file” in the top right while in the directory for the correct class. Select “Create new file” and name your file “midtermX/”. Obviously, replace the X with the correct midterm number. Adding the forward slash makes Git identify your “file” as a folder. Then, name the file “.gitkeep” in the new box that appeared after you pressed /. This file is just temporary.



Add a proper commit message and start a pull request by pressing “Propose changes”. This will create a new branch that you’re free to edit. Navigate to this branch and enter your newly created folder. Click “Add file” and upload your worksheet and solutions, each with .tex and .pdf files. This you can commit directly to the repository, since this is a branch owned by you. To avoid clutter, you also need to delete the .gitkeep file that you created before. Click on the file and select the three dots in the top right corner. Select “Delete file” and commit your changes.



Now your branch is ready to be reviewed and merged into main. Remember that you created a pull request already, so you just need to wait for a Student Services Chair to review your pull request. Congratulations, you've helped Student Services improve the resources available to the ECE community at Illinois!